APM1110_FA8_Valdez_ChanceJayden.pdf

https://github.com/ChanceValdez/APM1110-FA8

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Assessment Items (30 Points)

1. (6 points) An urn contains 10 red, 8 black, and 7 green balls. One of the colors is chosen at random (meaning that the chosen color is equally likely to be any of the 3 colors), and then 4 balls are randomly chosen from the urn. Let X be the number of these balls that are of the chosen color.

(a) (2 points) Find $P(X = 0)$.

Let C be a chosen color.
By the law of total probability,

$$P(X=0) = \sum_{c \in \{\text{red, black, green}\}} P(X=0 \mid C=c)\,P(C=c)$$

Given $C = c$ & $X = 0$, means all 4 balls are not of the selected color, c.
Let's find the total # of equally likely draws: $\binom{10+8+7}{4} = \binom{25}{4} = 12650$ draws.

Now find the probability of a ball to not be of a certain color:

$$P(X=0 \mid \text{red}) = \frac{\binom{15}{4}}{12650} = \frac{1365}{12650}$$

$$P(X=0 \mid \text{black}) = \frac{\binom{8}{4}}{12650} = \frac{2380}{12650}$$

$$P(X=0 \mid \text{green}) = \frac{\binom{7}{4}}{12650} = \frac{3060}{12650}$$

Weighing each color by weight:

$$P(X=0) = \frac{1}{3}\left[\frac{1365 + 2380 + 3060}{12650}\right] = \frac{1361}{7590} \approx 0.1793$$

(b) (2 points) Let X_i equal 1 if the ith ball selected is of the chosen color, and let it equal 0 otherwise. Find

$$P(X_i = 1), \quad i = 1, 2, 3, 4.$$

By sym. the ith ball is a uniformly random ball from the urn, so given a color $P(X_i = 1 \mid C = c)$ is that color's fraction of the 25 balls.

$$P(X_i = 1) = \frac{1}{3}\left[\frac{10+8+7}{25}\right] = \frac{1}{3}$$

(c) (2 points) Find $E[X]$.

Hint: Express X in terms of X_1, X_2, X_3, and X_4.

Since $X = X_1 + X_2 + X_3 + X_4$, linearity gives

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4]$$
$$= \frac{4}{3} \approx 1.3333$$

2. (8 points) The random variable X is said to have the Yule-Simon distribution if

$$P(X = n) = \frac{4}{n(n+1)(n+2)}, \quad n \geq 1.$$

(a) (3 points) Show that the preceding is actually a probability mass function. That is, show that

$$\sum_{n=1}^{\infty} P(X = n) = 1.$$

Hint: For (a), first use that

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right],$$

then use that

$$\frac{k}{n(n+k)} = \frac{1}{n} - \frac{1}{n+k}.$$

A Probability Mass Function requires the ff.:
1. the values must add up to 1 &
2. every value must be non-negative, $P(x) \geq 0$.

Let's start by substituting

$$\sum_{n=1}^{\infty} P(X=n) = \sum_{n=1}^{\infty} \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$$

Using the hint:

$$4\sum_{n=1}^{\infty} \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

$$2\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Let $S_N = \sum_{n=1}^{N} P(X=n)$

Then, plugging in the partial sums:

$$S_N = 2\left[\left(\frac{1}{2} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{12}\right) + \left(\frac{1}{12} - \frac{1}{20}\right) + \cdots + \left(\frac{1}{N(N+1)} - \frac{1}{(N+1)(N+2)}\right)\right]$$

$$S_N = 2\left[\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right]$$

Since an inf. series is the limit of its partial sums as $N \to \infty$:

$$\sum_{n=1}^{\infty} P(X=n) = \lim_{N\to\infty} S_n = \lim_{N\to\infty} 2\left[\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right] = 1$$

For nonnegativity: Since $n \geq 1$ implies that: $\frac{4}{n(n+1)(n+2)} > 0$

Since we proved that:

$$\sum_{n=1}^{\infty} \frac{4}{n(n+1)(n+2)} = \lim_{N\to\infty} 2\left[\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right] = 1$$

$\therefore P(X=n) \geq 0$ for all n & that the total sum equals 1, the Yule-Simon dist. is a PMF.

(b) (2 points) Show that

$$E[X] = 2.$$

To find $E[X]$ we use the def. for disc. rando. var.:

$$E[X] = \sum_{n=1}^{\infty} n \cdot P(X=n)$$

$$\sum_{n=1}^{\infty} n \cdot P(X=n) = \sum_{n=1}^{\infty} n \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$$

Using PFD:

$$\frac{1}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2}$$

$$1 = A(n+2) + B(n+1)$$

$$A = 1; \; B = -1$$

Using the result:

$$4\sum_{n=1}^{\infty} \frac{1}{n+1} - \frac{1}{n+2}$$

Let $S_N = \sum_{n=1}^{N} n \cdot P(X=n)$

Then plugging in partial sums:

$$S_N = 4\left[\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{N+1} - \frac{1}{N+2}\right)\right]$$

$$S_N = 4\left[\frac{1}{2} - \frac{1}{N+2}\right]$$

Since an inf. series is the limit of its partial sums as $N \to \infty$:

$$\sum_{n=1}^{\infty} n \cdot P(X=n) = \lim_{N\to\infty} S_N = \lim_{N\to\infty} 4\left[\frac{1}{2} - \frac{1}{N+2}\right] = 2.$$

Since we proved that:

$$\sum_{n=1}^{\infty} \frac{4}{(n+1)(n+2)} = \lim_{N\to\infty} 4\left[\frac{1}{2} - \frac{1}{N+2}\right] = 2$$

$\therefore$ the expected value $(E[X])$ of the Yule-Simon dist. is 2.

(c) (3 points) Show that

$$E[X^2] = \infty.$$

To find $E[X^2]$, we use:

$$E[X^2] = n^2 \cdot P(X=n)$$

$$\sum_{n=1}^{\infty} n^2 \cdot P(X=n) = \sum_{n=1}^{\infty} \frac{4n}{(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{n}{n^2+3n+2}$$

Let's use the LCT which states that:

The Limit Comparison Test Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

Let: $a_n = \frac{n}{n^2+3n+2}$ &

$b_n = \frac{1}{n}$ (a harmonic ser. which divs.)

$$\lim_{n\to\infty} \frac{\frac{n}{n^2+3n+2}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{n^2+3n+2}$$

L.H.R

$$\Rightarrow \lim_{n\to\infty} \frac{2n}{2n+3}$$

L.H.R

$$\Rightarrow \lim_{n\to\infty} \frac{2}{2} = 1$$

Since $c = 1$ is finite & greater than 0 and $\sum \frac{1}{n}$ diverges (harmonic ser.), $\sum a_n$ diverges.

$\therefore E[X^2] = 4\sum a_n = \infty.$

3. (4 points) The probability density function of X is given by

$$f(x) = \begin{cases} a + bx^2, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $E[X] = \dfrac{3}{5}$, find the values of a and b.

We are given the ff.:
- The PDF of X is $f(x) = a + bx^2$ in the range $[0,1]$ & 0 otherwise.
- The expected value, $E(X) = 3/5$.

We start with the integral of f over $[0,1]$ to satisfy the cond. for a PDF.

$$\int_0^1 a + bx^2 \, dx$$

Let's solve the integral first:

$$\int_0^1 a + bx^2 \, dx = \left(ax + \frac{1}{3}bx^3\right)\Big|_0^1$$
$$= \left(a(1) + \frac{1}{3}b(1)^3\right) - \left(a(0) + \frac{1}{3}b(0)^3\right)$$
$$= a + \frac{b}{3}$$

Since this a PDF, we set it equal to 1

$$a + \frac{b}{3} = 1$$

Now using the def. of $E[X]$ for cont. variables.

$$E[X] = \int_0^1 x(a + bx^2)\,dx = \int_0^1 ax + bx^3 \, dx$$

Solving the integral:

$$\int_0^1 ax + bx^3 \, dx = \left(\frac{1}{2}ax^2 + \frac{1}{4}bx^4\right)\Big|_0^1$$
$$= \left(\frac{1}{2}a(1)^2 + \frac{1}{4}b(1)^4\right) - \left(\frac{1}{2}a(0)^2 + \frac{1}{4}b(0)^4\right)$$
$$= \frac{a}{2} + \frac{b}{4}$$

Based on our given, we set the eqn. equal to $\frac{3}{5}$:

$$\frac{a}{2} + \frac{b}{4} = \frac{3}{5}$$

Now we can use subs./elim. Let's use the former.

1) $a + \frac{b}{3} = 1$ 2) $\frac{a}{2} + \frac{b}{4} = \frac{3}{5}$

From eqn. 1: $a = 1 - \frac{b}{3}$. Subs. into eqn 2:

$$\frac{1 - \frac{b}{3}}{2} + \frac{b}{4} = \frac{3}{5}$$
$$\frac{\frac{3-b}{3}}{2} + \frac{b}{4} = \frac{3}{5}$$
$$\frac{3-b}{6} + \frac{b}{4} = \frac{3}{5}$$
$$\frac{12 - 4b + 6b}{24} = \frac{3}{5}$$
$$\frac{12 + 2b}{24} = \frac{3}{5}$$
$$60 + 10b = 72$$
$$10b = 12$$
$$b = \frac{6}{5}$$

$$a = 1 - \frac{\frac{6}{5}}{3} = 1 - \frac{6}{15} = \frac{3}{5}$$

The values of a and b are $a = \frac{3}{5}$ & $b = \frac{6}{5}$. Confirmed via checking as well. Also $f(x) = \frac{3}{5} + \frac{\frac{6}{5}}{4} \ge 0$ on $[0,1]$ so its a valid PDF.

Checking:

eqn. 1: $\frac{3}{5} + \frac{\frac{6}{5}}{3} = \frac{3}{5} + \frac{6}{15} = \frac{45+30}{75} = \frac{75}{75} = 1$ ✓

eqn. 2: $\frac{\frac{3}{5}}{2} + \frac{\frac{6}{5}}{4} = \frac{3}{10} + \frac{6}{20} = \frac{60+60}{200} = \frac{120}{200} = \frac{3}{5}$ ✓

4. (4 points) Let X be a random variable having expected value μ and variance σ^2. Find the expected value and variance of

$$Y = \frac{X - \mu}{\sigma}.$$

Starting with our definitions:
- Mean = $\mu = E[X]$
- Variance = $\sigma^2 = \text{Var}(X) = E[(X - \mu)^2]$

Let's rewrite in the form $aX + b$

$$Y = \frac{X - \mu}{\sigma} \Rightarrow Y = \frac{X}{\sigma} - \frac{\mu}{\sigma} \Rightarrow Y = \frac{1}{\sigma}X - \frac{\mu}{\sigma}$$

Using the Linear Transformation Rule for $E[Y]$.

$$E[Y] = E[aX + b] = aE(X) + b$$
$$E\left[\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right] = \frac{1}{\sigma}E[X] - \frac{\mu}{\sigma} = \frac{1}{\sigma}\mu - \frac{\mu}{\sigma} = 0$$

Using the Linear Transformation Rule for $\text{Var}(Y)$

$$\text{Var}(Y) = \text{Var}(aX + b) = a^2\text{Var}(X)$$
$$\text{Var}\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \left(\frac{1}{\sigma}\right)^2\text{Var}(X) = \frac{1}{\sigma^2}\sigma^2 = 1$$

5. (8 points) Let X and Y be independent Poisson random variables with parameters $\lambda_1 = 2$ and $\lambda_2 = 3$, respectively. Let $W = X + Y$.

(a) (3 points) Using the property of sums of independent random variables, derive the moment generating function (MGF) $M_W(t)$ of W.

If Z is a Poisson random variable with parameter λ_i

$$M_Z(t) = E[e^{tZ}]$$

$$= \sum_{n=0}^{\infty} \frac{e^{tn} e^{-\lambda_i} \lambda^n}{n!}$$

$$= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda_i e^t)^n}{n!}$$

$$= e^{-\lambda_i} e^{\lambda_i e^t} = e^{\lambda_i (e^t - 1)} \text{ or } \exp\{(\lambda)(e^t - 1)\}$$

Substituting $\lambda_1 = 2$ & $\lambda_2 = 3$:

$P(X = 2) = e^{2(e^t - 1)}$

$P(Y = 3) = e^{3(e^t - 1)}$

$M_W(t) = M_X(t) + M_Y(t)$

$= e^{2(e^t - 1)} \cdot e^{3(e^t - 1)}$

$= e^{5(e^t - 1)}$ or $\exp\{(5)(e^t - 1)\}$

(b) (2 points) Based on the uniqueness property of MGFs, identify the probability distribution and parameter(s) of W.

The uniqueness property states that if $M_W(t)$ exists and is finite in some region about $t = 0$ then the distribution of W is uniquely determined.

Since $M_W(t) = \exp\{(5)(e^t - 1)\}$, then W is unambiguously a Poisson distribution.

Comparing $M_Z(t)$ & $M_W(t)$:

$M_Z(t) = \exp\{(\lambda)(e^t - 1)\}$

$M_W(t) = \exp\{(5)(e^t - 1)\}$

with $\lambda = 5$.

(c) (3 points) Determine the expected value $E[W]$ and the second moment $E[W^2]$ by successively differentiating $M_W(t)$ and evaluating it at $t = 0$.

Let's differentiate $M_W(t)$.

$$\frac{d}{dt} \exp\{(5)(e^t - 1)\} = \frac{d}{d} e^{5e^t - 5}$$

Using Chain Rule $\frac{d}{dx} e^u = e^u \cdot u'$

$= (5e^t)(e^{5e^t - 5})$

Plugging in $t = 0$ to get $E[W]$.

$= (5(1)) \cdot (e^{5e^{(0)} - 5})$

$= 5 \cdot 1 = 5$

Plugging in $t = 0$ to get $E[W^2]$

$25(1)(e^{5(1) - 5}) + (5(1))(e^{5(1) - 5})$

$= 25 + 5 = 30$

Let's check: A mean of Poisson is equal to its variance.

$\text{Var}(W) = E[W^2] - (E[W])^2$

$= 30 - 5^2 = 5 = E[W]$

Let's find the second derivative (using product rule).

$$\frac{d}{dt}(5e^t)(e^{5e^t - 5}) = (5e^t)(5e^t)(e^{5e^t - 5}) + (5e^t)(e^{5e^t - 5}) = 25e^{t^2}(e^{5e^t - 5}) + (5e^t)(e^{5e^t - 5})$$

$u'v + uv'$ or $uv' + u'v$